

Literature Status for arXiv:1601.06159: Numerical-Range Spectral Constants

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Status

This is a scoped `literature_already_answered` packet. It records that one concrete challenge in Michel Crouzeix’s *Some constants related to Numerical Ranges*, arXiv:1601.06159, namely to sharply improve the universal upper bound 11.08, is answered by Crouzeix–Palencia, *The numerical range is a $(1 + \sqrt{2})$ -spectral set*, SIAM J. Matrix Anal. Appl. 38 (2017), 649–655.

Source Question

The source paper defines best constants Q and Q_{cb} for scalar and complete polynomial functional calculus over numerical ranges, with

$$2 \leq Q \leq Q_{cb} \leq 11.08.$$

On PDF page 2, in the paragraph headed “Some open problems,” it says that the conjectures $Q = 2$ and $Q_{cb} = 2$, or determining the exact values, are difficult open problems. It then identifies a more modest challenge: sharply improve the upper bound 11.08. The same paragraph also asks whether $Q = Q_{cb}$.

Supporting Answer

The 2017 Crouzeix–Palencia paper proves the required upper-bound improvement. Its official SIAM abstract states that the numerical range of a Hilbert-space operator is a complete $(1 + \sqrt{2})$ -spectral set. In the notation of arXiv:1601.06159, this gives

$$Q_{cb} \leq 1 + \sqrt{2} \quad \text{and hence} \quad Q \leq 1 + \sqrt{2},$$

which sharply improves 11.08. The supporting paper explicitly cites Crouzeix’s 2016 source paper in its references.

Scope

This packet does not claim a new proof and does not claim that the full Crouzeix conjecture is solved. The exact conjectural constants $Q = 2$ and $Q_{cb} = 2$, the equality question $Q = Q_{cb}$, the finite-dimensional questions about $Q(3)$ and $Q_{cb}(3)$, and the other domain-constant questions in the source remain outside this scoped literature-status result.

The supporting publisher PDF was not directly available during this run: the SIAM PDF endpoint returned an access/challenge page rather than a PDF. The official DOI page and abstract provide the decisive bibliographic and theorem statement used here.

Search Evidence

The lane-1 queue selected arXiv:1601.06159. Cheap run indexes were searched for the arXiv id, source title, “numerical ranges”, “Davis-Wielandt”, “Crawford”, and “numerical index”. Existing adjacent packets on numerical index and Crawford-number questions were found, but no exact packet for this source paper or this upper-bound-improvement challenge was found before this packet was created. Web search for Crouzeix–Palencia and $1 + \sqrt{2}$ located the official SIAM page for DOI 10.1137/17M1116672.

References

- [1] M. Crouzeix, *Some constants related to Numerical Ranges*, arXiv:1601.06159; SIAM J. Matrix Anal. Appl. **37**, 420–442 (2016).
- [2] M. Crouzeix and C. Palencia, *The numerical range is a $(1 + \sqrt{2})$ -spectral set*, SIAM J. Matrix Anal. Appl. **38**, 649–655 (2017), doi:10.1137/17M1116672.